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Control Unit

Function

Gets the 8 bits instruction and emits the suitable control signals for each block. It decides the stalls from the branches and the interrupts.

The FSM is RUN in normal (00), when getting interrupt it turns to SAVE_PC (01) which saves the current pc to the memory and the current flags in the status register, when getting pc_saved it turns to LOAD_INTERRUPT (10) which loads the interrupt into the PC for only one cycle then goes to RUN_INTERRUPT (11) till gets an RTI instruction then goes to RUN.

Inputs

- rst
- Interrupt
- pc_saved
- Instruction[7:0]

Outputs

- write_reg_en
- write_reg_address_sel (0 means ra, 1 means rb)
- write_reg_data_sel (0 means from ALU, 1 means from Memory)
- read_reg_a_sel (0 means ra, 1 means R3)
- read_reg_b_sel (0 means rb, 1 means R3)
- push (in case of push instructions)
- pull (in case of pull instructions)
- flush_1_instrucion (LDM, LDD, STD)
- flush_2_instruciones (LOOP, CALL conditional branches)
- flush_3_instructions (RET, RTI, interrupt)
- push_pc
- pop_pc
- current_next_pc_sel (when 1 current (interrupt) and when 0 next (normal))
- pc_saving (high in the case of PC_SAVE)
- alu_op[3:0]
- alu_B_sel[1:0] (0 means da, 1 means db, 2 means immediate, 3 means INPUT)
- flag_en
- flag_address[1:0]
- branch_flag[1:0] (0 if using the flag out from CCR, 1 if using the direct zero flag from the ALU (LOOP), 2 means 1 (always jump))
- store_flags (from CCR to status)
- load_flags (from status to CCR)

- pc_load_en
- pc_load_data_sel (0 means db, 1 means ISR)
- mem_wr_en
- mem_interface_sel[1:0] (0 means SP, 1 means instruction, 2 means interrupt_instruction, 3 means data)
- write_to_reg (high when the instruction is writing into reg)
- destination_addr[1:0] (if the instruction is writing into reg, this is its address)
- use_memory (high when the instruction uses the memory)
- out_en (enable for the output instruction)

ALU_OP

- ADD = 0
- SUB = 1
- AND = 2
- OR = 3
- RLC B = 4
- RRC B = 5
- SETC = 6
- CLRC = 7
- NOT B = 8
- NEG B = 9
- INC B = 10
- DEC B = 11
- bypass B = 12

NOP

All enables are off and nothing changes in anything.

MOV

- write_reg_en = 1
- write_reg_address_sel = 0
- write_reg_data_sel = 0
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0
- flush_2_instrucions = 0
- flush_3_instructions = 0
- push_pc = 0

- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = Bypass B
- alu_B_sel[1:0] = 1
- flag_en = 0
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 0
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 1
- write_to_reg = 1
- destination_addr[1:0] = ra
- use_memory = 0
- out_en = 0

ADD

- write_reg_en = 1
- write_reg_address_sel = 0
- write_reg_data_sel = 0
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0
- flush_2_instrucions = 0
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = ADD
- alu_B_sel[1:0] = 1
- flag_en = 1
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0

- load_flags = 0
- pc_load_en = 0
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 1
- write_to_reg = 1
- destination_addr[1:0] = ra
- use_memory = 0
- out_en = 0

SUB

- write_reg_en = 1
- write_reg_address_sel = 0
- write_reg_data_sel = 0
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0
- flush_2_instrucons = 0
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = SUB
- alu_B_sel[1:0] = 1
- flag_en = 1
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 0
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 1
- write_to_reg = 1
- destination_addr[1:0] = ra
- use_memory = 0
- out_en = 0

AND

- write_reg_en = 1
- write_reg_address_sel = 0
- write_reg_data_sel = 0
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0
- flush_2_instrucons = 0
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = AND
- alu_B_sel[1:0] = 1
- flag_en = 1
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 0
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 1
- write_to_reg = 1
- destination_addr[1:0] = ra
- use_memory = 0
- out_en = 0

OR

- write_reg_en = 1
- write_reg_address_sel = 0
- write_reg_data_sel = 0
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0

- flush_2_instrucons = 0
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = OR
- alu_B_sel[1:0] = 1
- flag_en = 1
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 0
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 1
- write_to_reg = 1
- destination_addr[1:0] = ra
- use_memory = 0
- out_en = 0

RLC

- write_reg_en = 1
- write_reg_address_sel = 1
- write_reg_data_sel = 0
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0
- flush_2_instrucons = 0
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = RLC
- alu_B_sel[1:0] = 1
- flag_en = 1

- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 0
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 1
- write_to_reg = 1
- destination_addr[1:0] = rb
- use_memory = 0
- out_en = 0

RRC

- write_reg_en = 1
- write_reg_address_sel = 1
- write_reg_data_sel = 0
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0
- flush_2_instrucions = 0
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = RRC
- alu_B_sel[1:0] = 1
- flag_en = 1
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 0
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 1
- write_to_reg = 1

- destination_addr[1:0] = rb
- use_memory = 0
- out_en = 0

SETC

- write_reg_en = 0
- write_reg_address_sel = 0
- write_reg_data_sel = 0
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0
- flush_2_instrucons = 0
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = SETC
- alu_B_sel[1:0] = 0
- flag_en = 1
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 0
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 1
- write_to_reg = 0
- destination_addr[1:0] = 0
- use_memory = 0
- out_en = 0

CLRC

- write_reg_en = 0
- write_reg_address_sel = 0
- write_reg_data_sel = 0
- read_reg_a_sel = 0

- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0
- flush_2_instrucions = 0
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = CLRC
- alu_B_sel[1:0] = 0
- flag_en = 1
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 0
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 1
- write_to_reg = 0
- destination_addr[1:0] = 0
- use_memory = 0
- out_en = 0

PUSH

- write_reg_en = 0
- write_reg_address_sel = 0
- write_reg_data_sel = 0
- read_reg_a_sel = 1
- read_reg_b_sel = 0
- push = 1
- pop = 0
- flush_1_instrucion = 0
- flush_2_instrucions = 0
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0

- pc_saving = 0
- alu_op[3:0] = Bypass B
- alu_B_sel[1:0] = 0
- flag_en = 0
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 0
- pc_load_data_sel = 0
- mem_wr_en = 1
- mem_interface_sel[1:0] = 0
- write_to_reg = 0
- destination_addr[1:0] = 0
- use_memory = 1
- out_en = 0

POP

- write_reg_en = 1
- write_reg_address_sel = 1
- write_reg_data_sel = 1
- read_reg_a_sel = 1
- read_reg_b_sel = 0
- push = 0
- pop = 1
- flush_1_instrucion = 0
- flush_2_instrucions = 0
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = Bypass B
- alu_B_sel[1:0] = 0
- flag_en = 0
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 0

- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 0
- write_to_reg = 1
- destination_addr[1:0] = rb
- use_memory = 1
- out_en = 0

OUT

- write_reg_en = 0
- write_reg_address_sel = 0
- write_reg_data_sel = 0
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0
- flush_2_instrucons = 0
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = Bypass B
- alu_B_sel[1:0] = 1
- flag_en = 0
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 0
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 1
- write_to_reg = 0
- destination_addr[1:0] = 0
- use_memory = 0
- out_en = 1

IN

- write_reg_en = 1
- write_reg_address_sel = 1
- write_reg_data_sel = 0
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0
- flush_2_instrucons = 0
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = Bypass B
- alu_B_sel[1:0] = 3
- flag_en = 0
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 0
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 1
- write_to_reg = 1
- destination_addr[1:0] = rb
- use_memory = 0
- out_en = 0

NOT

- write_reg_en = 1
- write_reg_address_sel = 1
- write_reg_data_sel = 0
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0

- flush_2_instrucons = 0
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = NOT
- alu_B_sel[1:0] = 1
- flag_en = 1
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 0
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 1
- write_to_reg = 1
- destination_addr[1:0] = rb
- use_memory = 0
- out_en = 0

NEG

- write_reg_en = 1
- write_reg_address_sel = 1
- write_reg_data_sel = 0
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0
- flush_2_instrucons = 0
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = NEG
- alu_B_sel[1:0] = 1
- flag_en = 1

- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 0
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 1
- write_to_reg = 1
- destination_addr[1:0] = rb
- use_memory = 0
- out_en = 0

INC

- write_reg_en = 1
- write_reg_address_sel = 1
- write_reg_data_sel = 0
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0
- flush_2_instrucions = 0
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = INC
- alu_B_sel[1:0] = 1
- flag_en = 1
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 0
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 1
- write_to_reg = 1

- destination_addr[1:0] = rb
- use_memory = 0
- out_en = 0

DEC

- write_reg_en = 1
- write_reg_address_sel = 1
- write_reg_data_sel = 0
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0
- flush_2_instrucions = 0
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = DEC
- alu_B_sel[1:0] = 1
- flag_en = 1
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 0
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 1
- write_to_reg = 1
- destination_addr[1:0] = rb
- use_memory = 0
- out_en = 0

JZ

- write_reg_en = 0
- write_reg_address_sel = 0
- write_reg_data_sel = 0
- read_reg_a_sel = 0

- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0
- flush_2_instrucons = 1
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = 0
- alu_B_sel[1:0] = 0
- flag_en = 0
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 1
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 1
- write_to_reg = 0
- destination_addr[1:0] = 0
- use_memory = 0
- out_en = 0

JN

- write_reg_en = 0
- write_reg_address_sel = 0
- write_reg_data_sel = 0
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0
- flush_2_instrucons = 1
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0

- pc_saving = 0
- alu_op[3:0] = 0
- alu_B_sel[1:0] = 0
- flag_en = 0
- flag_address[1:0] = 1
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 1
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 1
- write_to_reg = 0
- destination_addr[1:0] = 0
- use_memory = 0
- out_en = 0

JC

- write_reg_en = 0
- write_reg_address_sel = 0
- write_reg_data_sel = 0
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0
- flush_2_instrucions = 1
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = 0
- alu_B_sel[1:0] = 0
- flag_en = 0
- flag_address[1:0] = 2
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 1

- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 1
- write_to_reg = 0
- destination_addr[1:0] = 0
- use_memory = 0
- out_en = 0

JV

- write_reg_en = 0
- write_reg_address_sel = 0
- write_reg_data_sel = 0
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0
- flush_2_instrucons = 1
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = 0
- alu_B_sel[1:0] = 0
- flag_en = 0
- flag_address[1:0] = 3
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 1
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 1
- write_to_reg = 0
- destination_addr[1:0] = 0
- use_memory = 0
- out_en = 0

LOOP

- write_reg_en = 1
- write_reg_address_sel = 0
- write_reg_data_sel = 0
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0
- flush_2_instrucons = 1
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = DEC
- alu_B_sel[1:0] = 0
- flag_en = 0
- flag_address[1:0] = 0
- branch_flag[1:0] = 1
- store_flags = 0
- load_flags = 0
- pc_load_en = 1
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 1
- write_to_reg = 1
- destination_addr[1:0] = ra
- use_memory = 0
- out_en = 0

JMP

- write_reg_en = 0
- write_reg_address_sel = 0
- write_reg_data_sel = 0
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0

- flush_2_instrucons = 1
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = 0
- alu_B_sel[1:0] = 0
- flag_en = 0
- flag_address[1:0] = 0
- branch_flag[1:0] = 2
- store_flags = 0
- load_flags = 0
- pc_load_en = 1
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 1
- write_to_reg = 0
- destination_addr[1:0] = 0
- use_memory = 0
- out_en = 0

CALL

- write_reg_en = 0
- write_reg_address_sel = 0
- write_reg_data_sel = 0
- read_reg_a_sel = 1
- read_reg_b_sel = 0
- push = 1
- pop = 0
- flush_1_instrucion = 0
- flush_2_instrucons = 1
- flush_3_instructions = 0
- push_pc = 1
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = Bypass B
- alu_B_sel[1:0] = 0
- flag_en = 0

- flag_address[1:0] = 0
- branch_flag[1:0] = 2
- store_flags = 0
- load_flags = 0
- pc_load_en = 1
- pc_load_data_sel = 0
- mem_wr_en = 1
- mem_interface_sel[1:0] = 0
- write_to_reg = 0
- destination_addr[1:0] = 0
- use_memory = 1
- out_en = 0

RET

- write_reg_en = 0
- write_reg_address_sel = 0
- write_reg_data_sel = 0
- read_reg_a_sel = 1
- read_reg_b_sel = 0
- push = 0
- pop = 1
- flush_1_instrucion = 0
- flush_2_instrucions = 0
- flush_3_instructions = 1
- push_pc = 0
- pop_pc = 1
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = Bypass B
- alu_B_sel[1:0] = 0
- flag_en = 0
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 0
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 0
- write_to_reg = 0

- destination_addr[1:0] = 0
- use_memory = 1
- out_en = 0

RTI

- write_reg_en = 0
- write_reg_address_sel = 0
- write_reg_data_sel = 0
- read_reg_a_sel = 1
- read_reg_b_sel = 0
- push = 0
- pop = 1
- flush_1_instrucion = 0
- flush_2_instrucions = 0
- flush_3_instructions = 1
- push_pc = 0
- pop_pc = 1
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = Bypass B
- alu_B_sel[1:0] = 0
- flag_en = 0
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 1
- pc_load_en = 0
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 0
- write_to_reg = 0
- destination_addr[1:0] = 0
- use_memory = 1
- out_en = 0

LDM

- write_reg_en = 1
- write_reg_address_sel = 1
- write_reg_data_sel = 0
- read_reg_a_sel = 0

- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 1
- flush_2_instrucons = 0
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = Bypass B
- alu_B_sel[1:0] = 2
- flag_en = 0
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 0
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 1
- write_to_reg = 1
- destination_addr[1:0] = rb
- use_memory = 0
- out_en = 0

LDD

- write_reg_en = 1
- write_reg_address_sel = 1
- write_reg_data_sel = 1
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 1
- flush_2_instrucons = 0
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0

- pc_saving = 0
- alu_op[3:0] = Bypass B
- alu_B_sel[1:0] = 2
- flag_en = 0
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 0
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 3
- write_to_reg = 1
- destination_addr[1:0] = rb
- use_memory = 1
- out_en = 0

STD

- write_reg_en = 0
- write_reg_address_sel = 0
- write_reg_data_sel = 0
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 1
- flush_2_instruciones = 0
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = Bypass B
- alu_B_sel[1:0] = 2
- flag_en = 0
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 0

- pc_load_data_sel = 0
- mem_wr_en = 1
- mem_interface_sel[1:0] = 3
- write_to_reg = 0
- destination_addr[1:0] = 0
- use_memory = 1
- out_en = 0

LDI

- write_reg_en = 1
- write_reg_address_sel = 1
- write_reg_data_sel = 1
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0
- flush_2_instrucons = 0
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = Bypass B
- alu_B_sel[1:0] = 0
- flag_en = 0
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 0
- pc_load_data_sel = 0
- mem_wr_en = 0
- mem_interface_sel[1:0] = 3
- write_to_reg = 1
- destination_addr[1:0] = rb
- use_memory = 1
- out_en = 0

STI

- write_reg_en = 0
- write_reg_address_sel = 0
- write_reg_data_sel = 0
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0
- flush_2_instrucons = 0
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = Bypass B
- alu_B_sel[1:0] = 0
- flag_en = 0
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 0
- pc_load_data_sel = 0
- mem_wr_en = 1
- mem_interface_sel[1:0] = 3
- write_to_reg = 0
- destination_addr[1:0] = 0
- use_memory = 1
- out_en = 0

SAVE_PC

- write_reg_en = 0
- write_reg_address_sel = 0
- write_reg_data_sel = 0
- read_reg_a_sel = 1
- read_reg_b_sel = 0
- push = 0
- pop = 1
- flush_1_instrucion = 0

- flush_2_instrucons = 0
- flush_3_instructions = 1
- push_pc = 0
- pop_pc = 1
- current_next_pc_sel = 1
- pc_saving = 1
- alu_op[3:0] = Bypass B
- alu_B_sel[1:0] = 0
- flag_en = 0
- flag_address[1:0] = 0
- branch_flag[1:0] = 0
- store_flags = 0
- load_flags = 0
- pc_load_en = 0
- pc_load_data_sel = 0
- mem_wr_en = 1
- mem_interface_sel[1:0] = 0
- write_to_reg = 0
- destination_addr[1:0] = 0
- use_memory = 1
- out_en = 0

LOAD_INTERRUPT

- write_reg_en = 0
- write_reg_address_sel = 0
- write_reg_data_sel = 0
- read_reg_a_sel = 0
- read_reg_b_sel = 0
- push = 0
- pop = 0
- flush_1_instrucion = 0
- flush_2_instrucons = 1
- flush_3_instructions = 0
- push_pc = 0
- pop_pc = 0
- current_next_pc_sel = 0
- pc_saving = 0
- alu_op[3:0] = 0
- alu_B_sel[1:0] = 0
- flag_en = 0

- flag_address[1:0] = 0
- branch_flag[1:0] = 2
- store_flags = 0
- load_flags = 0
- pc_load_en = 1
- pc_load_data_sel = 1
- mem_wr_en = 0
- mem_interface_sel[1:0] = 2
- write_to_reg = 0
- destination_addr[1:0] = 0
- use_memory = 0
- out_en = 0

RUN_INTERRUPT

All the instructions are like the normal mode except that the memory interface selector in the parts where the instruction doesn't use memory = 2 instead of 1.

You can implement RTI only in RUN_INTERRUPT or in the two modes it won't matter.